

Redox Reactions & Galvanic Cells

Study Notes aligned to Chapters 8 & 9

Chapter 8 Redox reactions (8.1–8.4) & Chapter 9 Galvanic cells (9.1–9.5): oxidation & reduction, oxidation numbers, half-equations, the reactivity series, galvanic cells, the electrochemical series, predicting reactions, everyday power sources & corrosion. *Electrolysis is Chapter 10 and is not covered here.*

Chapter 8 — Redox reactions

8.1 Oxidation and reduction

8.2 Oxidation numbers

8.3 More complex redox equations

8.4 The reactivity series of metals

Chapter 9 — Galvanic cells

9.1 Galvanic cells

9.2 The electrochemical series

9.3 Predicting direct redox reactions

9.4 Everyday sources of power

9.5 Corrosion

Reference

R1 Exam traps & glossary

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CHAPTER 8 — REDOX REACTIONS

8.1 Oxidation and Reduction

A **redox reaction** transfers electrons between species. **Oxidation** and **reduction** always occur together — electrons lost by one species are gained by another.

OIL RIG

Oxidation Is Loss · Reduction Is Gain (of electrons)

THREE WAYS TO DESCRIBE REDOX (HISTORICAL → MODERN)

Process	Electrons	Oxygen	Hydrogen	Ox. number
Oxidation	lost	gained	lost	increases
Reduction	gained	lost	gained	decreases

Early definitions were based on gain/loss of **oxygen** (combustion, metal extraction); the modern, general definition is **electron transfer**.

Other examples of redox: combustion, corrosion, metal displacement, cellular respiration and photosynthesis, and metals reacting with acids are all electron-transfer reactions.

Oxidising agents and reducing agents

DEFINITIONS

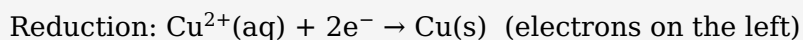
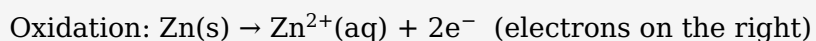
- **Oxidising agent (oxidant):** accepts electrons → is itself **reduced**.
- **Reducing agent (reductant):** donates electrons → is itself **oxidised**.

COMMON CONFUSION

The oxidising agent is the species that gets **reduced**; the reducing agent is the species that gets **oxidised**.

Writing simple half-equations

A **half-equation** shows one process with its electrons. For simple metal/ion systems, add electrons to balance the charge:



CHECK

Both atoms **and** charge must balance. The number of electrons equals the change in total charge across the half-equation.

8.2 Oxidation Numbers

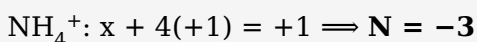
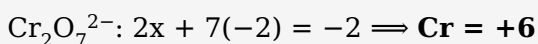
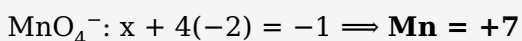
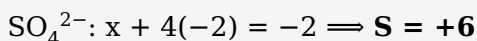
The **oxidation number** is the charge an atom *would* have if all bonds were fully ionic — the tool for tracking electron transfer.

OXIDATION NUMBER RULES (PRIORITY ORDER)

1. Free element = **0** (Na, O₂, Cl₂, S₈, P₄).
2. Monatomic ion = its charge (Na⁺ = +1, S²⁻ = -2, Al³⁺ = +3).
3. Group 1 = +1; Group 2 = +2; Al = +3 in compounds.
4. F = -1 always. H = +1 (but -1 in metal hydrides, e.g. NaH).
5. O = -2 (but -1 in peroxides e.g. H₂O₂; +2 in OF₂).
6. Sum = **0** in a neutral compound; = **ion charge** in a polyatomic ion.

Calculating oxidation numbers

WORKED EXAMPLES



Using oxidation numbers to name chemicals (Stock names)

For metals with more than one common charge, the oxidation number is shown as a **Roman numeral** in the name:

Formula	Oxidation no.	Name
FeCl ₂	Fe = +2	iron(II) chloride
FeCl ₃	Fe = +3	iron(III) chloride
Cu ₂ O	Cu = +1	copper(I) oxide
CuO	Cu = +2	copper(II) oxide

Using oxidation numbers to identify oxidation & conjugate pairs

RULES

- Oxidation number **increases** → oxidised; **decreases** → reduced; **no change** → not redox.
- Each oxidant/reductant forms a **conjugate redox pair**: oxidised form + $n e^- \rightleftharpoons$ reduced form (e.g. Cu^{2+}/Cu).

Disproportionation

One element is simultaneously oxidised and reduced:

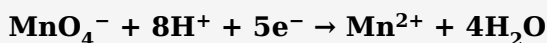


8.3 More Complex Redox Equations

BALANCING A HALF-EQUATION IN ACIDIC SOLUTION – 5 STEPS

1. Balance atoms except O and H.
2. Balance **O** with **H₂O**.
3. Balance **H** with **H⁺**.
4. Balance **charge** with **e⁻**.
5. Check atoms and charge.

EXAMPLE – PERMANGANATE REDUCED IN ACID



BASIC SOLUTION

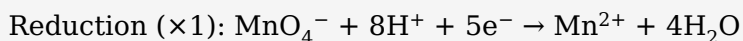
Balance as if acidic, then add **OH⁻ to both sides** equal to the H⁺; combine $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ and cancel duplicate water.

Combining half-equations into a full equation

METHOD

Multiply each half-equation so the **electrons are equal**, add them, then cancel electrons and any identical species.

EXAMPLE – MnO_4^- OXIDISING Fe^{2+} IN ACID



Charge check: left +17 = right +17 ✓

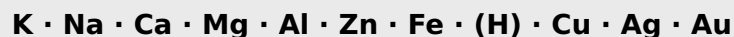
APPLICATION – THE DICHROMATE BREATHALYSER

Acidified orange dichromate oxidises ethanol in breath; chromium is reduced from +6 (orange $\text{Cr}_2\text{O}_7^{2-}$) to +3 (green Cr^{3+}). The orange→green colour change shows alcohol was present.

8.4 The Reactivity Series of Metals

Metals are **reducing agents** — they lose electrons to form positive ions. The more reactive a metal, the more readily it is oxidised, i.e. the **stronger a reducing agent** it is.

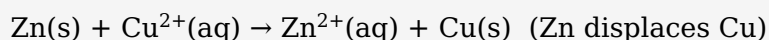
REACTIVITY (ACTIVITY) SERIES — MOST → LEAST REACTIVE



Metals above hydrogen displace H_2 from acids; the most reactive also react with water. This order is the reverse of the metals' positions in the electrochemical series (§9.2).

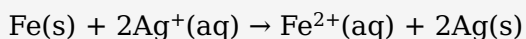
Metal displacement reactions

A **more reactive metal displaces the ion of a less reactive metal** from solution (the reactive metal is oxidised; the less-reactive ion is reduced):



PREDICTING A DISPLACEMENT

Will iron displace silver? Fe is more reactive than Ag, so **yes**:



The reverse ($\text{Ag} + \text{Fe}^{2+}$) does **not** occur.

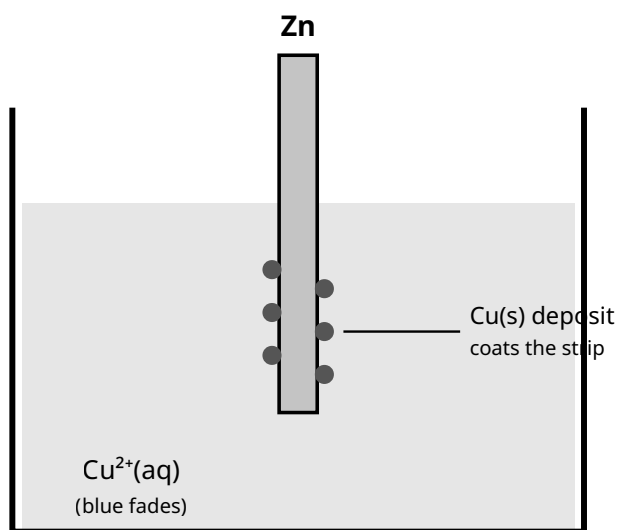


Figure 1. Metal displacement: a zinc strip in copper(II) sulfate. Zn (more reactive) is oxidised and dissolves ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$); Cu^{2+} is reduced to copper metal that coats the strip ($\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$). Overall $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$; the blue solution fades as Cu^{2+} is consumed.

Reactions of metals with water, acids and oxygen

Reactivity	With cold water	With dilute acid
Very reactive (K, Na, Ca)	react readily $\rightarrow$ metal hydroxide + H_2	violent $\rightarrow$ salt + H_2
Moderate (Mg, Al, Zn, Fe)	slow / only with steam	steady $\rightarrow$ salt + H_2
Unreactive (Cu, Ag, Au)	no reaction	no reaction (below H)

e.g. $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$ · $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

CHAPTER 9 — GALVANIC CELLS

9.1 Galvanic Cells

A **galvanic (voltaic) cell** converts chemical energy into electrical energy using a **spontaneous** redox reaction. Separating the two half-reactions forces electrons through an external wire — the current.

Half-cells

Each electrode + its solution is a **half-cell** (one oxidation, one reduction). If a half-cell has no solid conductor (e.g. $\text{Fe}^{3+}/\text{Fe}^{2+}$) an **inert electrode** (Pt or graphite) is used.

Electrode	Process	Galvanic polarity
Anode	Oxidation	negative (–)
Cathode	Reduction	positive (+)

MEMORY AIDS & MOVEMENT

- **AN OX** = ANode OXidation · **RED CAT** = REDuction CAThode.
- Electrons flow **anode → cathode** in the wire.
- In the salt bridge: **anions → anode, cations → cathode**.

Purpose of the salt bridge

The salt bridge (inert electrolyte, e.g. KNO_3) **completes the circuit** and **maintains electrical neutrality** in each half-cell by allowing ions to migrate. It does *not* carry electrons.

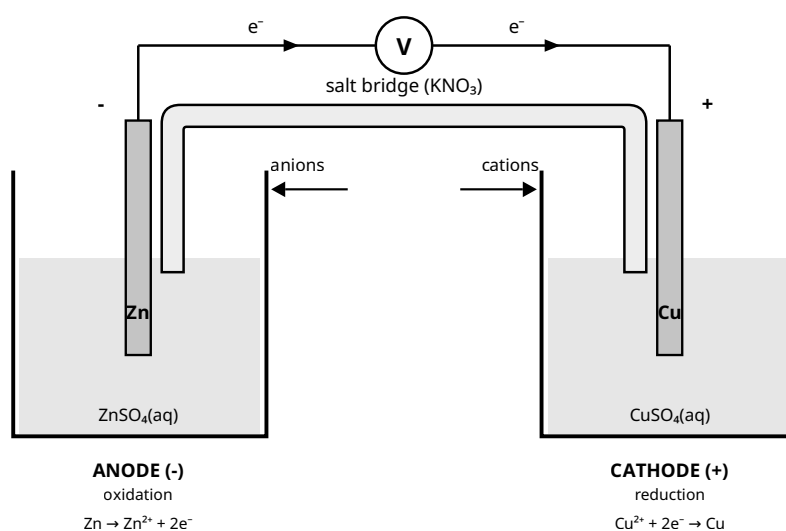
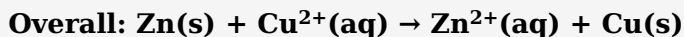
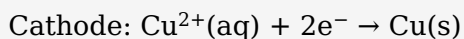
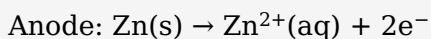


Figure 2. Galvanic (Daniell) cell. Electrons flow anode (–) → cathode (+); the salt bridge keeps each half-cell neutral (anions → anode, cations → cathode).

Writing an overall equation for a cell reaction

DANIELL CELL



Shorthand cell notation

Anode on the left. | = phase boundary; || = salt bridge.



9.2 The Electrochemical Series

Half-cell potentials are measured against the **Standard Hydrogen Electrode (SHE)**, defined as exactly **0.00 V**. The ordered list of standard reduction potentials is the **electrochemical series**.

STANDARD CONDITIONS (THE "°")

Solutions **1.0 mol L⁻¹** · gases **100 kPa** · **25 °C** · measured vs the SHE.

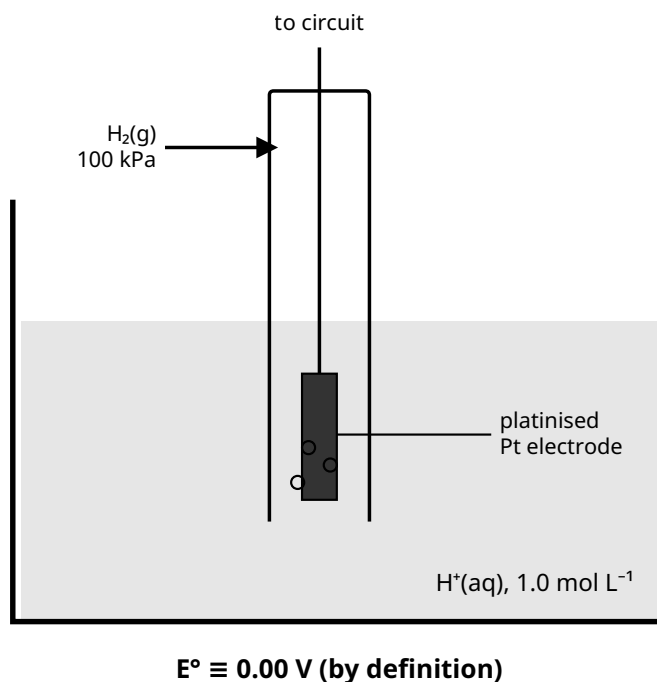


Figure 3. The Standard Hydrogen Electrode (SHE): H₂ at 100 kPa over inert Pt in 1.0 mol L⁻¹ H⁺ at 25 °C, assigned E° = 0.00 V so all other potentials can be measured against it.

Standard electrode potentials

READING THE SERIES (ALL VALUES ARE REDUCTIONS)

- **More positive E°** → stronger **oxidising agent** (top-left; e.g. F₂, MnO₄⁻).
- **More negative E°** → its reduced form is a stronger **reducing agent** (bottom-right; e.g. Li, K, Na).

Predicting cell reactions & calculating the voltage

CELL EMF

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

Both values taken as reductions from the series; the half-cell with the higher E° is the cathode (reduced). A positive E°_{cell} confirms a spontaneous cell reaction.

EXAMPLE — ZN | CU CELL

$$E^{\circ}_{\text{cell}} = E^{\circ}(\text{Cu}^{2+}/\text{Cu}) - E^{\circ}(\text{Zn}^{2+}/\text{Zn}) = (+0.34) - (-0.76) = +1.10 \text{ V}$$

E° DOES NOT SCALE WITH AMOUNT

E° is **intensive** — when you multiply a half-equation to balance electrons, do **not** multiply its E° value.

9.3 Predicting Direct Redox Reactions

The electrochemical series predicts whether two species react **directly** (mixed together, not in a cell).

THE RULE

A species reacts if the **oxidant (left side) has a higher E° than the reductant (right side)** it is mixed with — i.e. the higher- E° species is reduced and the lower- E° species is oxidised.

Equivalent test: $E^{\circ}_{\text{cell}} = E^{\circ}_{\text{oxidant}} - E^{\circ}_{\text{reductant}} > 0 \Rightarrow$ reaction occurs.

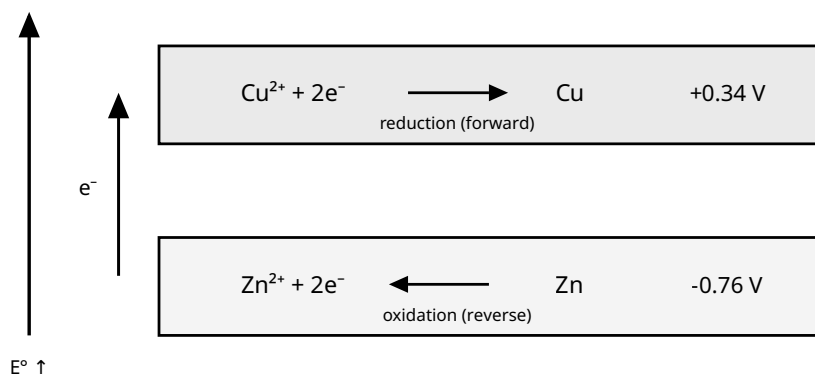
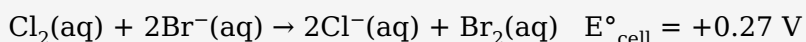


Figure 4. Predicting a reaction from the electrochemical series (the "anticlockwise rule"). The couple with the higher E° (top) undergoes reduction; the lower- E° couple (bottom) is reversed to oxidation. Electrons pass from the reductant (Zn) up to the oxidant (Cu^{2+}), so $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ is spontaneous.

WORKED EXAMPLE — WILL Cl_2 OXIDISE Br^- ?

Cl_2/Cl^- (+1.36 V) is higher than Br_2/Br^- (+1.09 V), so Cl_2 is reduced and Br^- is oxidised:



(The reverse — $\text{Br}_2 + \text{Cl}^-$ — does not occur.)

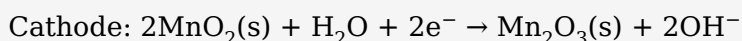
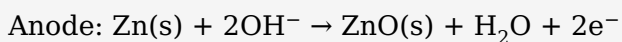
Limitations of predictions

- Valid only at **standard conditions**; concentration, temperature, pressure shift the real potential.
- Predicts **feasibility, not rate** — a spontaneous reaction may be very slow.
- Values assume aqueous solutions; different solvents/complexes can change behaviour.

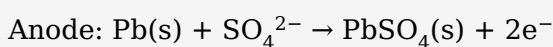
9.4 Everyday Sources of Power

Type	Feature	Examples
Primary	Non-rechargeable; reaction not reversible	dry (Leclanché) cell, alkaline cell
Secondary	Rechargeable; reaction reversed by an external supply	lead-acid accumulator, lithium-ion cell
Fuel cell	Reactants supplied continuously; not stored in the cell	hydrogen-oxygen fuel cell

ALKALINE CELL (PRIMARY)



LEAD-ACID ACCUMULATOR (SECONDARY)



Recharging drives these reactions in reverse (this is electrolysis — Chapter 10).

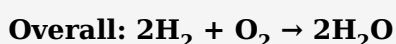
LITHIUM-ION CELL (SECONDARY)

Rechargeable cell in phones, laptops and EVs. Li^+ ions shuttle between a graphite anode and a lithium-metal-oxide cathode (e.g. LiCoO_2) during discharge/charge:



High energy density and light weight; uses a lithium-salt organic (non-aqueous) electrolyte.

HYDROGEN-OXYGEN FUEL CELL (ACIDIC)



Because chemical energy is converted **directly** to electricity (not via heat/combustion), fuel cells are highly efficient and produce only water. Challenges: producing H_2 cleanly and storing it safely (compressed, cryogenic, or in metal hydrides).

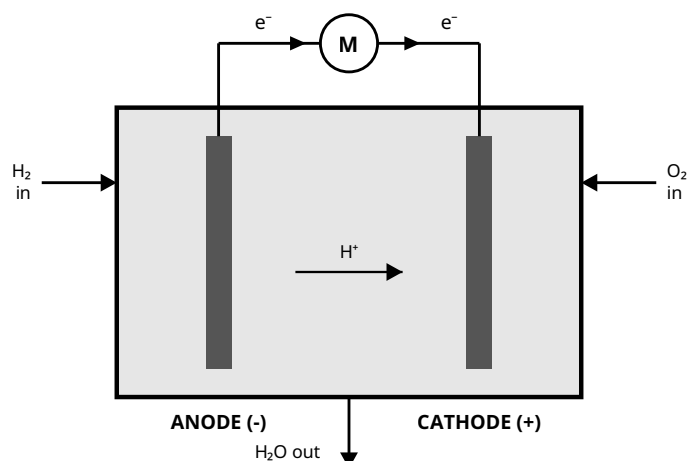
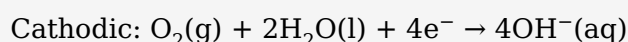
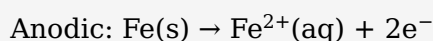


Figure 5. Hydrogen-oxygen fuel cell. Anode: $2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$; cathode: $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$. Electrons flow anode $\rightarrow$ cathode through the load (M); H^+ migrates through the electrolyte. Overall $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ — the only product is water.

9.5 Corrosion

Corrosion is the unwanted oxidation of a metal. **Wet corrosion** (rusting of iron) is electrochemical and needs **both water and oxygen**; it is faster with dissolved salts or acids.

WET CORROSION PROCESS



Fe^{2+} is further oxidised by O_2 to hydrated iron(III) oxide $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ = rust.

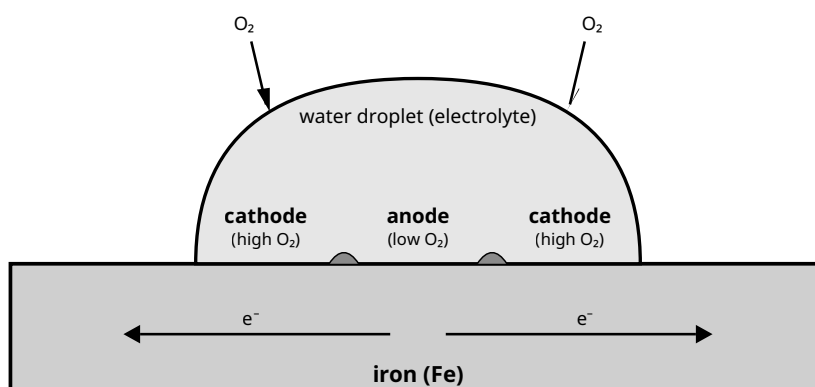


Figure 6. Wet corrosion under a water droplet. Centre (low O_2) = anode (Fe oxidised); edges (high O_2) = cathode (O_2 reduced). Fe^{2+} and OH^- meet to deposit rust.

Electrochemical protection

Method	How it works
Sacrificial anode (Zn/Mg block)	A more reactive metal (more negative E°) is attached and oxidised <i>instead</i> of the iron; the iron becomes the cathode. Used on ship hulls/pipelines.
Cathodic protection (impressed current)	An external supply forces the iron to be the cathode so it cannot oxidise.

Galvanising (Zn coat)	Barrier <i>plus</i> sacrificial protection — Zn oxidises before Fe even if scratched.
Barrier (paint/oil)	Excludes water and O ₂ ; fails if scratched.
Alloying (stainless steel)	Cr/Ni form a self-repairing protective oxide.

WHY SACRIFICIAL PROTECTION WORKS

A metal with a **more negative E°** than iron (Zn -0.76 , Mg -2.37 vs Fe -0.44) is a stronger reducing agent, so it is oxidised preferentially and protects the iron.

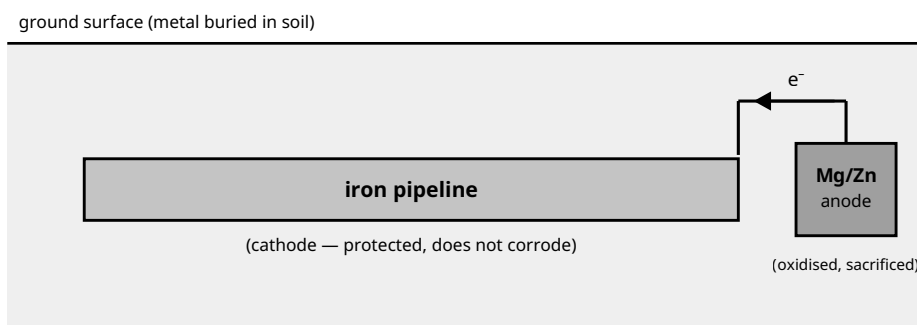


Figure 7. Cathodic (sacrificial) protection. A more reactive metal (Mg or Zn) is connected to the buried iron. The block is the anode and is oxidised ($\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$); its electrons flow into the iron, forcing the iron to be the cathode so it cannot corrode. The sacrificial anode is consumed and replaced periodically.

REFERENCE

R1 Exam Traps & Glossary

TOP TRAPS (CHAPTERS 8-9)

- Oxidising agent is **reduced**; reducing agent is **oxidised**.
- Never change E° when multiplying a half-equation (E° is intensive).
- Salt bridge maintains **neutrality/completes the circuit** — it does not carry electrons.
- Always **check charge balance** in half-equations.
- A positive E°_{cell} means **feasible**, not fast.
- More reactive metal = stronger reducing agent = **more negative E°**.

Glossary

Term	Definition
Redox reaction	Electron-transfer reaction (simultaneous oxidation + reduction).
Oxidation / Reduction	Loss / gain of electrons (ox. number increases / decreases).
Oxidising / Reducing agent	Electron acceptor (itself reduced) / donor (itself oxidised).
Oxidation number	Charge an atom would have if all bonds were ionic.
Half-equation	Shows one process (oxidation or reduction) with its electrons.
Conjugate redox pair	Oxidised and reduced forms linked by e^- transfer.
Reactivity series	Metals ordered by ease of oxidation (reducing strength).
Displacement reaction	A more reactive metal reduces the ion of a less reactive metal.
Anode / Cathode	Electrode of oxidation / reduction.
Salt bridge	Maintains neutrality and completes a galvanic circuit.

E° (standard potential)	Reduction potential of a half-cell vs the SHE at standard conditions.
SHE	Standard Hydrogen Electrode; reference defined as 0.00 V.
Electrochemical series	Half-reactions ordered by E° (oxidising strength).
Primary / Secondary cell	Non-rechargeable / rechargeable galvanic cell.
Fuel cell	Galvanic cell with continuously supplied reactants.
Corrosion	Unwanted oxidation of a metal by its environment.

R2 Standard Reduction Potentials (25 °C)

All half-equations written as reductions. Strongest oxidant at top (most positive E°); strongest reductant at bottom (most negative E°). Always use the exact values on your official exam data sheet.

Ordered strongest oxidant (top) → strongest reductant (bottom)

Reduction half-equation	E° (V)	Note
$\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-$	+2.87	strongest oxidant
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.77	
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+1.51	acidified permanganate
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+1.36	
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+1.33	acidified dichromate
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.23	
$\text{Br}_2(\text{l}) + 2\text{e}^- \rightarrow 2\text{Br}^-$	+1.09	
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.80	
$\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$	+0.77	
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$	+0.40	
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34	
$\text{Sn}^{4+} + 2\text{e}^- \rightarrow \text{Sn}^{2+}$	+0.15	
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00	SHE reference
$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13	
$\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$	-0.14	
$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.25	
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.44	
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76	used in galvanising
$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-$	-0.83	
$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}(\text{s})$	-1.66	
$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.37	
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71	
$\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca}(\text{s})$	-2.87	
$\text{K}^+ + \text{e}^- \rightarrow \text{K}(\text{s})$	-2.93	
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04	strongest reductant

KEY RELATIONSHIP

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ (positive $\Rightarrow$ spontaneous galvanic reaction). $\Delta G^\circ = -nFE^\circ_{\text{cell}}$, $F = 96\,500 \text{ C mol}^{-1}$.

Aligned to Pearson Chemistry WA 12, Chapters 8 & 9. Electrolysis (Chapter 10) is covered separately. Content written as original study notes.